Department of Computer Science	Final Exam for *Logic and Sets*
Vrije Universiteit Amsterdam	April 1, 2022, 12:15–14:30

This exam has 4 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

1. Island of liars and truth speakers (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet three islanders A, B and C.
A says: "B or C is a truth speaker, but not both."
B says: "A is a truth speaker."
C says: "B is a truth speaker."

You need to determine, *by means of a truth table*, which of these three islanders speak the truth and which ones lie.

Solution:

t_A	t_B	t_C	$t_B \oplus t_C$	$t_A \leftrightarrow (t_B \oplus t_C)$	$t_B \leftrightarrow t_A$	$t_C \leftrightarrow t_B$
T	T	T	F	F	T	T
T	T	F	T	T	T	F
T	F	T	T	T	F	F
T	F	F	F	F	F	T
F	T	T	F	T	F	T
F	T	F	T	F	F	F
F	F	T	T	F	T	F
F	F	F	F	T	T	T

Only in the last row the last three columns contain T. So A, B and C are all liars.

2. CNF (8 points)

Build a formula in CNF that corresponds to the following truth table (using the method explained in the third logic lecture).

p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	F

Solution: $(\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge (p \vee q \vee \neg r) \wedge (p \vee q \vee r)$

3. **Logic circuit** (10 points)
Give a logic circuit, using only AND, OR and NOT gates, that represents $x \to y$.

Solution: The easiest solution is to build a circuit for $\neg x \vee y$.

```
x ---> NOT ---
             |
             v
y ---------> OR --->
```

A slighly more complicated solution is to build a circuit for $\neg(x \wedge \neg y)$.

```
x ---------> AND ---> NOT --->
              ^
              |
y ---> NOT ----
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4. **Predicate logic** (6+9 points)
Consider the two formulas $\exists x\, C(x)$ and $C(y)$.

   a) Which of the two formulas semantically entails the other? Motivate your answer.
   Solution: The second formula semantically entails the first one. If $C(y)$ holds on a model, it means there is an interpretation of the free variable y in this model for which C holds. This same interpretation can be used for x in the existential quantification of the first formula to show that this formula holds on this model too.

   b) Give a model to show that the two formulas are not semantically equivalent.
   Solution: Let $A = \{a, b\}$, let C hold only or a, and let $y = b$. Then the first formula holds on the model, because $C(a)$ means there is an x such that $C(x)$ holds. But the second formula does not hold on the model, because $C(b)$ does not hold.

5. **Sets** (9 points)
Consider the following set-theoretic equality

$$(A \backslash B) \cup ((B \cup C') \backslash B) \; = \; ((A' \cap C) \cup B)'.$$

Prove it using the rules of the algebra of sets.

Solution:

$$\begin{aligned}
(A \backslash B) \cup ((B \cup C') \backslash B) &= \\
(A \cap B') \cup ((B \cup C') \cap B') &= \\
(A \cap B') \cup ((B \cap B') \cup (C' \cap B')) &= \\
(A \cap B') \cup (\emptyset \cup (C' \cap B')) &= \\
(A \cap B') \cup (C' \cap B') &= \\
(A \cup C') \cap B' &= \\
((A \cup C')' \cup B)' &= \\
((A' \cap C) \cup B)'
\end{aligned}$$

6. Relations (3+5+4 points)
In the set $V = \{1, 2, 3, 4\}$ consider the relation R given by the set of pairs

$$\{\langle 1,1\rangle, \langle 1,2\rangle, \langle 1,3\rangle, \langle 1,4\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 3,3\rangle, \langle 4,4\rangle\}.$$

a) Show that R is not an equivalence relation.

Solution: R is not an equivalence relation since R is not symmetric. For example, $\langle 1,2\rangle \in R$ but $\langle 2,1\rangle \notin R$.

b) Show that R is an ordering relation which is not total.

Solution: R is reflexive ($\langle 1,1\rangle$,$\langle 2,2\rangle$,$\langle 3,3\rangle$,$\langle 4,4\rangle \in R$). Furtermore, R is transitive and anti-symmetric (with regard to the ordering 1<2,2<3,2<4). R is not total since $\langle 3,4\rangle, \langle 4,3\rangle \notin R$.

c) Determine the minimal and maximal elements of R. Does there exist a largest element? Does there exist a smallest element?

Solution: 1 is the only minimal element and hence also the smallest element. 3 and 4 are maximal elements and hence there is no largest element.

7. Functions (6+6 points)
We are given the functions $root : \mathbb{R} \to \mathbb{R}$, $exp : \mathbb{R} \to \mathbb{R}$, and $add : \mathbb{R} \to \mathbb{R}$, defined by

$$root(x) \ = \ \sqrt{x}, \quad exp(x) \ = \ e^x, \quad \text{and} \quad add(x) \ = \ x+1.$$

a) For each of these functions check whether it is total and whether it is surjective.

Solution: *root* is not total and not surjective; *exp* is total and not surjective; *add* is total and surjective.

b) Give a description (an expression for $f(x)$) for the function $f := root \circ add^{-1} \circ exp$ and specify the domain of definition D_f and the range R_f.

Solution: $f(x) = (root \circ add^{-1})(e^x) = root(e^x - 1) = \sqrt{e^x - 1}$,
$y = f(x) \geq 0 \Leftrightarrow e^x - 1 \geq 0 \Leftrightarrow e^x \geq 1 \Leftrightarrow x \geq 0$ gives $D_f = \{x \in \mathbb{R} : x \geq 0\}$,
$R_f = \{y \in \mathbb{R} : y \geq 0\}$.

8. Induction and recursion (3+9 points)
Consider the sequence $(t_n)_{n\in\mathbb{N}}$ of numbers defined recursively by

$$t_1 := 0, \qquad t_{n+1} := t_n + \frac{1}{n(n+1)} + 1.$$

We claim that the following statement is true for all natural numbers n:

$$t_n = n - \frac{1}{n}.$$

a) Verify by explicit computation that the claim is true for $n = 1$, $n = 2$ and $n = 3$.
Solution: $t_1 = 0 = 1-1$, $t_2 = 0+\frac{1}{2}+1 = 1\frac{1}{2} = 2-\frac{1}{2}$, $t_3 = 1\frac{1}{2}+\frac{1}{6}+1 = 2\frac{2}{3} = 3-\frac{1}{3}$.

b) Prove by mathematical induction that the statement holds true for all natural numbers n.
Solution: Base case: $t_1 = 0 = 1 - 1$.
Induction step: Let $m \geq 1$. Assume that $t_m = m - \frac{1}{m}$ (IH). Then we have

$$t_{m+1} = t_m + \frac{1}{m(m+1)} + 1 \overset{\text{(IH)}}{=} m - \frac{1}{m} + \frac{1}{m(m+1)} + 1$$
$$(m+1) - \frac{1}{m+1} = m - \frac{1}{m} + \frac{1}{m(m+1)} + 1$$

Hence (IH) implies $t_{m+1} = (m+1) - \dfrac{1}{m+1}$.